

Assignment 9

This exercise sheet is dedicated to quadrature, focusing on the order of a quadrature rule and on composite rules.

Graded exercise

Exercise 31 (Composite trapezoidal rule)

- (a) Write a PYTHON function `composite_trapezoidalquad` that computes the approximation to $\int_a^b f(x)dx$ for a given function $f : [a, b] \rightarrow \mathbb{R}$ using the composite trapezoidal rule on equispaced intervals and without using `scipy` routines. In input, the function should take an handle to the function f to be integrated, the two extremes a and b of the integration interval and the number N of subintervals for the composite rule. The output should be the approximation to the desired integral.
- (b) We now perform a convergence study for the quadrature rule implemented in the previous task. For this, we consider the function $f(x) = x \sin(2\pi x)$ on $[0, 1]$. The exact value for the integral is given by $\int_0^1 x \sin(2\pi x)dx = -\frac{1}{2\pi}$. Write a PYTHON script that computes the absolute error between the result of the composite rule from the previous task and the exact integral, for $N = 2^i$ with $i = 1, \dots, 8$. Produce a double logarithmic (loglog) plot of the error versus $h = \frac{1}{N}$ (don't forget to hand it in!). Which order of convergence do you observe?

So far we have considered integration over an interval, so in dimension $d = 1$. Assume now that we want to approximate a multidimensional integral

$$\int_{[0,1]^d} f(\mathbf{x})d\mathbf{x},$$

with $d \in \mathbb{N}$, $d > 1$. To do this, we take the composite trapezoidal rule and we *tensorize*. Namely, if we use $N + 1$ quadrature nodes in each direction, our quadrature nodes in $[0, 1]^d$ will be the $M := (N + 1)^d$ possible combinations for $(x_1, \dots, x_d)$, where each x_i , $i = 1, \dots, d$, is a node of the one-dimensional quadrature rule. We see that the number

of function evaluations grows *exponentially* with the dimension d . For the composite trapezoidal rule, the quadrature error can still be bounded as

$$|I - \tilde{I}| \leq Ch^2, \quad (1)$$

for a constant C depending on the integrand, and now h is the largest diameter of the hypercubes in the subdivision induced by the tensorization.

- (c) Assuming we use a uniform subdivision (all intervals in each direction have the same size), rewrite the estimate (1) in terms of the number M of function evaluations required by the multidimensional quadrature rule. How does the convergence rate *with respect to* M behave as $d \rightarrow \infty$?

Remark: This phenomenon is known as *curse of dimensionality*. Want to learn more and how to overcome it? Then attend the course Monte Carlo Methods in the Master's programme!

Other exercises

Exercise 32 (Maximal order of a quadrature rule)

To approximate $I(f) = \int_0^1 f(x)dx$ for a continuously differentiable function f , we use an approximation of the kind $\tilde{I}(f) = \alpha_0 f(0) + \alpha_1 f'(0) + \beta_0 f(x_0)$, with $\alpha_0, \alpha_1, \beta_0 \in \mathbb{R}$ and $x_0 \in [0, 1]$.

Estimate which is the maximal order that such a quadrature rule can have, for a proper choice of $\alpha_0, \alpha_1, \beta_0$ and x_0 , and estimate these parameters so that the maximal order is attained.

Exercise 33 (Quadrature for a weighted integral)

We consider the weighted integral $I_w(f) := \int_0^1 w(x)f(x)dx$, with weight $w(x) = \sqrt{x}$. We approximate $I_w(f)$ with a quadrature formula of the kind $\tilde{I}(f) = af(c_1)$, for $a \in \mathbb{R}$ and $c_1 \in [0, 1]$. Determine a and c_1 such that the quadrature rule is of the highest order p possible, and determine such p .